Exponential sample complexity for an analytic activation arbitrarily close to tanh

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Abstract

For every fixed integer $r \geq 1$ and every $\eta > 0$, we construct one computable odd, strictly increasing, real-analytic activation ϕ with range $(-1, 1)$, global Lipschitz constant one, $\phi'(0) = 1$, strict concavity on the positive half-line, and $\|\phi - \tanh\|_{C^r(\mathbb{R})} < \eta$. For this fixed activation, the original Gaussian Bayesian classifier requires exponentially many examples in teacher depth on a uniform Boolean cube. Teachers have depth $L_d = Dd + O(1)$, width and coefficient bound $O(d)$, and constant global output margin. At every fixed Gaussian exponent $\gamma \geq 1$, every sample of size at most $\exp(cd)$ fails the posterior confidence criterion at every sufficiently large width $N \geq \exp(C_\gamma d)$. The critical endpoint uses finite-width entropy certificates and a parallel-circuit normal form for paths near their scalar baseline. A fixed-width cold-limit statement also holds. An appendix gives a second activation with every derivative globally bounded and retains the $\exp(\Omega(L))$ lower bound.

The construction is computationally contrived. It encodes computable hard truth tables in tiny analytic bumps, and decodes them by finite differences that cancel the tanh background. No efficient evaluator for the activation is asserted. This is a theorem about specially constructed activations arbitrarily close to tanh, not about tanh itself; it does not settle tanh, ReLU, or GELU.

Status. This conventional mathematical research proof has received independent AI-agent reviews of the activation, decoder, capacity, energy, and posterior arguments, including the critical finite-width target-mass and canonical-state arguments. It is not Lean verified or human peer reviewed.

1 Model and theorem

Let $X_d = \{-1, 1\}^d$ and $h^0(x) = (1, x)$. A common-width-N network of weighted depth L is

$$h^s(x) = \phi(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_W(x) = v^\mathsf{T} h^{L-1}(x).$$

There are no separate hidden biases. Write $W^L = v^\mathsf{T}$ and $P_F(W) = \prod_{s=1}^{L} \|W^s\|_F$. The independent Gaussian entry variances are t^2/d in the first matrix, t^2/N internally, and t^2/N^2 in the readout. Set $\sigma^2 = t^2/N$. The energy is

$$\kappa_N(W) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2. \tag{1}$$

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In energy coordinates the prior is isotropic $N(0, \sigma^2 I)$. For a consistently labeled sample $S = ((x_i, y_i))_{i=1}^s$, condition only on

$$E_S = \{|z_W(x)| > 1/2 \ (x \in X_d), \ y_i z_W(x_i) > 0 \ (1 \le i \le s)\}. \tag{2}$$

Call the posterior $Q_{S,\sigma}$ and define

$$e_{S,\sigma}(f) = 2^{-d} \big|\{x : Q_{S,\sigma}(f(x) z_W(x) > 0) \le 2/3\}\big|.$$

Here $\|g\|_{C^r(\mathbb{R})} = \max_{0 \le j \le r} \sup_{u \in \mathbb{R}} |g^{(j)}(u)|$.

Theorem 1.1. *Fix $r \ge 1$ and $\eta > 0$. There exist one fixed computable activation ϕ, a computable family $f_d : X_d \to \{-1, 1\}$, and constants $c > 0, d_0, D, K$ with the following properties.*

(i) ϕ is odd, real analytic, strictly increasing, and globally 1-Lipschitz, with range $(-1, 1)$ and $\phi'(0) = 1$. It is strictly concave on $(0, \infty)$ and $\|\phi - \tanh\|_{C^r(\mathbb{R})} < \eta$.

(ii) The targets have teachers of weighted depth $L_d = Dd + K$, width and coefficient bound $C(d+1)$, and global signed output margin greater than one. The coefficients can be chosen with bit lengths polynomial in d. One may take $D = \max(16, r + 2) + 1$.

(iii) For every $\zeta > 0$ there is C_ζ such that for $d \ge d_0$, all $N \ge \exp(C_\zeta d)$, all $0 < \sigma^2 \le N^{-(1+\zeta)}$, and every sample of at most $\exp(cd)$ inputs labeled by f_d,

$$e_{S,\sigma}(f_d) \ge 139/1024 > 1/8.$$

(iv) For each $1 \le n \le \exp(cd)$, every fixed finite $N \ge M(d, n)$ for a polynomial M has the same conclusion for all sufficiently small $\sigma > 0$, uniformly over all samples of at most n inputs.

(v) There is C_{crit} such that the same every-sample conclusion holds at the actual critical scale $\sigma^2 = 1/N$ for every $N \ge \exp(C_{\mathrm{crit}} d)$ and $d \ge d_0$.

Constants may depend on the fixed r, η. The targets are evaluated under the uniform input law.

Thus the number of samples is exponential in L_d. The finite-scale width threshold is polynomial in the maximum permitted sample budget, but is not asserted polynomial in every smaller actual sample size. The coefficient and architectural teacher resources are polynomial in depth. The hard information is in the one fixed activation. The constants may depend on a fixed Gaussian exponent; no uniform claim for exponents approaching one with the problem resources is implicit in the theorem.

2 Universal capacity and computable hard tables

The tables must be chosen independently of the activation that later encodes them. We use a capacity bound allowing different activations at different neurons.

Proposition 2.1 (Universal finite-circuit capacity)**.** *There are universal constants A, b such that any Boolean function with a globally unit-margin network of Frobenius product P has a bounded-fan-in Boolean circuit of size at most $A(4 + \lceil P \rceil)^b$. The network may have arbitrary finite depth and widths, and each neuron may use its own zero-preserving 1-Lipschitz activation. The Boolean depth can be $O(\log^2(4 + P))$; a threshold circuit has depth $O(\log(4 + P))$.*

A proof is given in Appendix A. Computability of the activations is unnecessary: the circuits are nonuniform and store finite tables as constants.

Lemma 2.2. *There are universal $a > 0, d_0$ and a computable family f_d such that for $d \geq d_0$, every globally unit-margin network in the class of Proposition 2.1 with product at most e^{ad} disagrees with f_d on at least $7/16$ of the uniform cube.*

Proof. Enumerate all bounded-fan-in Boolean circuits on d inputs with at most $\lfloor 2^{d/4} \rfloor$ gates. There are $\exp(O(d2^{d/4}))$ such circuits. A Hamming ball of relative radius $q = 7/16$ among 2^d-bit truth tables has at most $2^{H_2(q)2^d}$ elements, with binary entropy $H_2(q) < 1$. The union covers fewer than 2^{2^d} tables eventually. Choose the lexicographically first table at distance at least $q2^d$ from every listed circuit. At smaller dimensions with no such table choose a constant. Finite exhaustive search computes this choice. Choose $a > 0$ so small that $A(4 + \lceil e^{ad} \rceil)^b \leq 2^{d/4}$ eventually, and apply Proposition 2.1. The choice refers only to Boolean circuits, not to the later activation. □

3 An analytic perturbation with weighted derivative control

Fix $M = \max(16, r + 2)$, $D = M + 1$, and $m = M + 3$. Let J be a sufficiently large fixed dyadic integer, increased from 2^{m+10} if needed below. Set

$$\Delta_d = 2^{-d-4}, \quad r_d = \Delta_d/J, \quad c_{d,k} = -4d - \tfrac{1}{4} - \Delta_d k, \qquad 0 \leq k < 2^d.$$

For a sufficiently small fixed positive rational τ, put $\delta_d = \tau r_d 2^{-Md} = \tau(16J)^{-1}2^{-Dd}$ and define

$$S(u) = \sum_{d \geq 1} \sum_{k < 2^d} \delta_d f_d(k)(u/c_{d,k})^3 \exp[-((u - c_{d,k})/r_d)^2], \tag{3}$$

$$R(u) = S(u) - S(-u), \qquad \phi(u) = \tanh u + R(u). \tag{4}$$

Lemma 3.1. *The function S is entire and computable. For every fixed $j \leq \max(r, 5)$,*

$$\sup_{\mathbb{R}} |S^{(j)}| \leq C_j \tau (16J)^{j-1} \sum_{d \geq 1} 2^{-(M+1-j)d}. \tag{5}$$

There are constants C, C_J such that

$$|R'(u)| \leq C\tau \min(u^2, 1) \operatorname{sech}^2 u, \tag{6}$$

$$|R''(u)| \leq C_J \tau |\tanh'' u|. \tag{7}$$

A sufficiently small τ makes ϕ satisfy all activation properties in Theorem 1.1.

Proof. On a complex compact set $|\Re z|, |\Im z| \leq A$, for all large d the real part of $(z - c_{d,k})^2$ is at least cd^2. The entire d-band is bounded by a polynomial factor times $2^d \delta_d \exp(-cd^2/r_d^2)$, a summable sequence. This proves local uniform convergence and entire analyticity. The explicit tails and computable finite tables also prove computability, without an efficiency assertion.

For one bump write $v = (u - c)/r$ and $\rho = r/c$. Its shape is $(1 + \rho v)^3 e^{-v^2}$ with $|\rho| \leq 1$. Its j-th derivative is bounded by $C_j r^{-j} e^{-v^2/2}$. The elementary lattice estimate

$$\sup_s \sum_{k \in \mathbb{Z}} e^{-(s - Jk)^2/2} \leq 3 \qquad (J \geq 4)$$

follows by choosing the nearest lattice point and summing the remaining tails. This gives (5).

For a weighted derivative bound use $\operatorname{sech}^2 u \ge e^{-2|u|}$, $|c_{d,k}| \le 4d + 5/16$, and $|u| \le |c_{d,k}| + r_d|v|$. The derivative's polynomial factor times $e^{-v^2+2r_d|v|}$ is at most $Ce^{-v^2/2}$, uniformly since $r_d \le 1/128$. Therefore

$$\frac{|S'(u)|}{\operatorname{sech}^2 u} \le C\tau \sum_{d\ge 1} e^{-(M\log 2-8)d} \le C'\tau.$$

A second derivative has an extra r_d^{-1}, giving

$$\frac{|S''(u)|}{\operatorname{sech}^2 u} \le CJ\tau \sum_{d\ge 1} e^{-((M-1)\log 2-8)d} \le C'_J\tau.$$

Both series converge because $M \ge 16$.

Near zero the cubic factor gives additional powers. For $|u| \le 1$, a single first derivative divided by u^2 is bounded by

$$\frac{\delta_d}{|c|^3}\left(3 + \frac{2|u||u-c|}{r_d^2}\right) e^{-(u-c)^2/r_d^2}.$$

Here $|u-c| \in [3d, 6d]$. Its band sum is at most $C\tau 2^{-(M-1)d}(r_d + r_d^{-1})e^{-9d^2/r_d^2}$. For a second derivative divided by u, the exact bracket is $\delta_d c^{-3}[6g(v) + 6ug'(v)/r_d + u^2 g''(v)/r_d^2]$, where $g(v) = e^{-v^2}$; its band sum is at most

$$C\tau 2^{-(M-1)d}(r_d + r_d^{-1} + r_d^{-3})e^{-9d^2/r_d^2}.$$

Both series are $O(\tau)$. Since $\operatorname{sech}^2 u$ is bounded below on $[-1, 1]$, this supplies the additional u^2 in (6). Also $|\tanh'' u|$ is bounded below by a positive constant times $|u|$ on this interval, and by a positive constant times $\operatorname{sech}^2 u$ outside it. Apply these bounds at u and $-u$ to get (6)–(7).

The perturbation is odd and vanishes to order three at zero, so $\phi(0) = 0$ and $\phi'(0) = 1$. Choose τ so both weighted constants are at most $\varepsilon < 1/10$. Then $\phi'(u) > 0$, while

$$\phi'(u) \le \operatorname{sech}^2 u + \varepsilon u^2 \operatorname{sech}^2 u \le \operatorname{sech}^2 u + \varepsilon \tanh^2 u \le 1,$$

using $|u| \le |\sinh u|$. The second weighted estimate gives $\phi''(u) < 0$ for $u > 0$. Uniform summability in (5) implies $S(u) \to 0$ as $|u| \to \infty$, so ϕ has limits ± 1 and range $(-1, 1)$. Shrink τ once more to obtain the required C^r distance. □

Two consequences used below, after decreasing τ if needed, are

$$|\phi(u) - u| \le 2u^2, \tag{8}$$

$$1 - \phi(u) \le Ce^{-2u} \quad (u \ge 0). \tag{9}$$

The first follows from $\phi'(0) = 1$ and a global second-derivative bound. The second follows by integrating $\phi' \le (1+\varepsilon)\operatorname{sech}^2$ to infinity. More precisely $1 - \phi(u)$ lies between $(1-\varepsilon)(1-\tanh u)$ and $(1+\varepsilon)(1-\tanh u)$.

4 Finite differences decode the hidden truth table

Put $h_d = \Delta_d/(2m)$ and define

$$F_d(u) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \phi(u + jh_d). \tag{10}$$

Lemma 4.1. *For all sufficiently large d, uniformly over $0 \le k < 2^d$,*

$$F_d(c_{d,k}) = \delta_d(f_d(k) + e_{d,k}), \qquad |e_{d,k}| \le 1/100.$$

Proof. The central bump has value exactly $\delta_d f_d(k)$. Every off-center query $u + jh_d$, $1 \le j \le m$, lies between h_d and $\Delta_d/2$ from this center, hence is at least h_d from all centers in its band. Gaussian-polynomial absorption and the lattice bound make its same-band contribution at most $C\delta_d \exp[-J^2/(16m^2)]$. Other same-band bumps at the central query contribute at most $C\delta_d e^{-J^2/4}$. The coefficient absolute sum is 2^m, so fixed sufficiently large J makes their aggregate error less than $\delta_d/1000$.

Other-band contributions divided by δ_d are bounded by a polynomial factor times

$$\sum_{e \ne d} 2^{e-D(e-d)} \exp[-9(e-d)^2 J^2 2^{2e+8}].$$

This tends to zero. For $e \le d/2$ its exponent dominates $-cd^2 4^e$; for $d/2 < e < d$ it dominates $-c4^{d/2}$; for $e > d$ it decreases faster still. All prefactors are dominated in these three splits. Reflected bumps have distance at least $3(d+e)$ from their negative-band centers, so satisfy a stronger estimate.

The tanh part is an m-th forward difference. Its iterated-integral formula bounds it by

$$h_d^m \sup_{v \in [u, u+mh_d]} |\tanh^{(m)} v| \le C_m \Delta_d^m e^{-8d}.$$

For $v \le -1$, the estimate $|\tanh^{(m)} v| \le C_m e^{2v}$ follows from its convergent series in e^{2v}. Dividing by $\delta_d = \tau(16J)^{-1}2^{-Dd}$ gives $O(2^{-2d}e^{-8d})$, since $m = M + 3$. The fixed number of query terms does not affect convergence. These bounds prove the claim. □

Let $\mathrm{code}(x) = \sum_{i=0}^{d-1} 2^i (x_{i+1}+1)/2$. The first teacher layer has $m+1$ neurons with preactivations $c_{d,\mathrm{code}(x)} + jh_d$, $0 \le j \le m$. Their biases are $O(d)$ and other coefficients are at most $1/64$. A second hidden neuron uses the binomial weights in (10). Its output has the desired sign and amplitude $\delta_d(1 + O(1/100))$.

Follow it by $Dd + K_0$ scalar hidden layers of weight two. Choose fixed K_0 so $2^{K_0}\tau/(16J)$ is a sufficiently small positive amplitude ρ, up to a factor two. By (8), cumulative relative error during the amplification is $O(\rho)$: the sum of ideal intermediate amplitudes is a geometric series bounded by 4ρ. Choose ρ to make that error less than $1/10$ and use readout $2/\rho$. The signed output margin exceeds one. Depth is $L_d = Dd + K_0 + 3$, active width is $m + 1$, and all coefficients are bounded by $C(d+1)$. First-layer coefficients are rational with $O(d)$ bits. Fixed remaining coefficients can be approximated with polynomial bit lengths while preserving the margin by layerwise Lipschitz propagation.

5 A low-energy memorizer in the exponential sample range

Choose an even integer $A \ge C_1 \log(n+2)$ large enough for (9), and put

$$h_p(x) = \tfrac{1}{2}\{\phi(A(1 - d + p \cdot x)) + \phi(A)\}.$$

The first argument is $A(1 - 2\,\mathrm{Ham}(x,p))$, with absolute value at least A. Oddness and saturation imply $|h_p(x) - \mathbf{1}_{\{x=p\}}| \le Ce^{-2A}$. If $p_1, \dots, p_s$, $s \le n$, are the distinct negative sample points, then

$$z_0(x) = 2 - 4\sum_{j=1}^{s} h_{p_j}(x)$$

is within $1/10$ of the ideal memorizer taking values ± 2. The constant feature is $\phi(A)$, with readout $2/\phi(A) - 2s$; each point-feature readout is -2. Thus its first Frobenius norm squared is at most $F^2 = A^2[1 + n(d+1)^2]$, and its readout squared norm is $O((n+1)^2)$.

Zero-pad to

$$M_0 = \lceil \max\{4, n+1, dF^2\} \rceil. \tag{11}$$

For $N \geq M_0$, clone every feature $\lfloor N/M_0 \rfloor$ times and divide its readout coefficients by that number. The output is unchanged, the first weighted energy is at most one, and the old readout v_0 has

$$B = N\|v_0\|_2^2 \leq B_0 \leq C(d+1)^3(n+1)^3 \log^2(n+2). \tag{12}$$

Reduce the hard exponent a from Lemma 2.2 so $a \leq 1/100$, and fix

$$q = \exp(a/(100D)) > 1.$$

Choose a fixed small $\alpha > 0$ with $\alpha \leq c_0(q-1)$ for a sufficiently small universal c_0. Put $\ell = L_d - 2$ and $\rho_1 = \alpha q^{1-\ell}$. The first scalar hidden matrix has every row $\rho_1 v_0^\top$; the remaining $\ell - 1$ scalar hidden matrices have every entry q/N. Their ideal states are

$$v_j(x) = \alpha q^{j-\ell} z_0(x).$$

Lemma 5.1. *For all sufficiently large d and $n \leq \exp(cd)$, where $c < a/1000$ is fixed, this construction followed by readout entries $1/(\alpha N)$ has global margin at least 1.5, fits every sample, and has energy at most*

$$K_d = 2 + (\ell - 1)q^2 + \alpha^{-2} \leq q^2 L_d + C_*. \tag{13}$$

The constants are independent of d, n, N.

Proof. Equation (8) gives a relative-error bootstrap. The first actual-to-ideal ratio differs from one by at most $C|v_1|$. Subsequently write $y_j = q y_{j-1}(1 + \theta_j)$ with $|\theta_j| \leq C|y_{j-1}|$. Under $|y_i/v_i| \leq 2$, the sum of these relative errors is at most $C'\alpha/(q-1)$. Our choice of α makes this less than $1/100$; the product of $1 + \theta_j$ closes the bootstrap and puts every ratio in $[9/10, 11/10]$. All preactivations stay in $[-1, 1]$. The final output therefore has margin at least $(9/10)(1.9) > 1.5$, with the signs of z_0.

From (12), $\log B_0 \leq 3cd + O(\log d)$, whereas $q^{-2\ell} = \exp[-ad/50 + O(1)]$. Thus $B\rho_1^2 \leq 1$ eventually, uniformly over the sample range. This is the first scalar energy. The other scalar energies are q^2 each and the readout energy is α^{-2}; adding the original first energy proves (13). □

The product of the block energies in (1) is $dP_F(W)^2$. Hence

$$dP_F(W)^2 \leq (\kappa_N(W)/L_d)^{L_d}. \tag{14}$$

For all large d a globally strict-margin-$1/2$ function with an energy-$K_d + 1$ representative therefore has a unit-margin representative of product at most

$$P_{\text{cold}} = \frac{2}{\sqrt{d}}((K_d + 1)/L_d)^{L_d/2} \leq C q^{L_d}/\sqrt{d}. \tag{15}$$

In the remaining proof write $L = L_d$, $K = K_d$, and $M = M(d, n) = M_0$.

6 Cold posterior concentration

Lemma 6.1. *For fixed d, n, L, N with $N \geq M(d, n)$,*

$$\sup_{|S| \leq n} Q_{S,\sigma}\{\kappa_N > K + 1\} \longrightarrow 0 \qquad (\sigma \downarrow 0),$$

where the supremum is over consistently labeled samples.

Proof. The memorizing witness has margin at least 1.5 and energy at most K. Continuity of ϕ and finiteness of X_d imply that a nonempty ball in energy coordinates lies inside E_S and has energy at most $K + 1/4$. If p is the parameter dimension, this ball has Gaussian probability at least $c_S \sigma^{-p} \exp[-(K + 1/4)/(2\sigma^2)]$. For $\eta = 1/[4(K + 1)]$, the Gaussian exponential moment gives

$$\mu_\sigma\{\kappa_N > K + 1\} \leq \eta^{-p/2} \exp[-(K + 3/4)/(2\sigma^2)].$$

The ratio tends to zero. Only finitely many distinct conditioning events arise from samples of at most n inputs on the finite cube, so convergence is uniform over those events. □

7 Actual finite Gaussian scales

The following argument uses only $\phi(0) = 0$ and $\mathrm{Lip}(\phi) \leq 1$. Normalize the input by $\sqrt{d}$, hidden states by $\sqrt{N}$, and use energy-coordinate blocks

$$A_1 = \sqrt{d/N}\, W^1, \quad A_s = W^s \ (2 \leq s < L), \quad A_L = \sqrt{N}\, v^\mathsf{T}.$$

The normalized hidden activation is $u \mapsto \phi(\sqrt{N}u)/\sqrt{N}$, still zero-preserving and 1-Lipschitz. The normalized input norm is at most $\sqrt{2}$.

Lemma 7.1 (Uniform perturbation)**.** *If $\kappa_N(U) \leq K + 1$ and each normalized block of B has operator norm at most η, then*

$$\sup_{x \in X_d} |z_{U+B}(x) - z_U(x)| \leq \sqrt{2} L \eta \left(\frac{K + 1 + 2\eta\sqrt{L(K+1)} + L\eta^2}{L - 1} \right)^{(L-1)/2}.$$

Define

$$A_* = \left(\frac{K + 2}{L - 1} \right)^{(L-1)/2}, \quad \eta = \min\left\{ 1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2} L A_*} \right\}. \tag{16}$$

Then the output perturbation is at most $1/8$.

Proof. Replace blocks one at a time. Zero preservation bounds intermediate state norms by the input norm times preceding operator norms. Lipschitzness bounds each replacement's effect by η times the other $L - 1$ factors. Write $a_s = \|U_s\|_F$; those factors are at most $a_s + \eta$. The sum of their squares, even including the omitted factor, is at most $K + 1 + 2\eta\sqrt{L(K+1)} + L\eta^2$. Arithmetic–geometric mean and the sum over replacements prove the bound. The second choice in (16) makes this numerator at most $K + 2$, and the last choice gives the error $1/8$. □

Set

$$b^2 = \max\{2(d + 1)\log 2 + 4\log 8,\ 16\log 9 + 8\log(8L)\}. \tag{17}$$

In energy coordinates let T be the set where the first and last blocks have Frobenius norm at most tb, and every internal block has operator norm at most tb. This set is closed, convex, and centrally symmetric. It has raw Gaussian mass at least $1/2$. Indeed the first and last bounds each fail with probability at most $1/8$ by the chi-square exponential moment. Two $1/4$-nets of the N-dimensional unit sphere have at most 9^{2N} pairs; the scalar Gaussian tail and a union bound give, for each internal block, failure probability at most

$$2\exp(2N\log 9 - Nb^2/8) \le 1/(4L).$$

The sum of all these failure bounds is at most $1/2$.

Proposition 7.2. *Suppose $tb \le \eta$ for (16)–(17). For every sample of at most n inputs, the posterior gives mass at least $1 - 2e^{-1/(2\sigma^2)}$ to Boolean functions admitting a global unit-margin representative of product at most*

$$P_{\mathrm{fin}} = \frac{8}{3\sqrt{d}}((K+1)/L)^{L/2} \le Cq^L/\sqrt{d}. \tag{18}$$

Proof. Let w_S be the memorizing witness. Lemma 7.1 gives $w_S + T \subset E_S$. The Gaussian shift formula and central symmetry of T imply

$$\mu_\sigma(E_S) \ge \mu_\sigma(w_S + T) \ge \tfrac{1}{2}\exp[-K/(2\sigma^2)].$$

Gaussian isoperimetry for a set of Gaussian mass at least $1/2$ gives

$$\mu_\sigma\{\mathrm{dist}(W,T)^2 > K+1\} \le \exp[-(K+1)/(2\sigma^2)].$$

Distance here is Euclidean distance in energy coordinates. Division bounds the posterior exceptional mass by $2e^{-1/(2\sigma^2)}$.

Outside this set let B be the metric projection of W onto T and set $U = W - B$. Then $\kappa_N(U) \le K+1$, and all normalized blocks of B have operator norm at most $tb \le \eta$. Lemma 7.1 makes z_U differ from z_W by at most $1/8$ everywhere. Their signs agree and $|z_U| > 3/8$. Multiply its readout by $8/3$, and apply (14) and (13), to obtain (18). □

Fix $\zeta > 0$. It suffices to require

$$N \ge \left\lceil \max\{M, 2, (b/\eta)^{2/\zeta}, 2\log 64\} \right\rceil. \tag{19}$$

Then $\sigma^2 \le N^{-(1+\zeta)}$ implies $t = \sqrt{N\sigma^2} \le N^{-\zeta/2}$, so $tb \le \eta$; the posterior exceptional mass is at most $1/32$. For $K \le q^2L + C_*$,

$$((K+2)/L)^L \le Cq^{2L}, \qquad A_*^2 \le e((K+2)/L)^L \le Cq^{2L}.$$

Thus $\eta^{-1} \le C\,\mathrm{poly}(L)q^L$. The bulk constant is polynomial in d, L, and $M \le \exp(Cd)$ uniformly for $n \le \exp(cd)$. Since $L = Dd + O(1)$, a bound $N \ge \exp(C_\zeta d)$ satisfies (19) throughout this sample range.

8 Completion at the cold and fixed-gap scales

Since $L_d \log q = ad/100 + O(1)$, both (15) and (18) are at most e^{ad} for all sufficiently large d. Lemma 2.2 gives error at least $7/16$ for every Boolean function in those posterior events. By the cold comparison or finite-scale projection argument, the event has posterior mass at least $31/32$.

The posterior Gibbs error is at least $(31/32)(7/16) = 217/512$. If $e = e_{S,\sigma}(f_d)$, the Gibbs error is at most $(1-e)/3 + e$. Hence

$$e \geq [3(217/512) - 1]/2 = 139/1024 > 1/8.$$

All estimates are uniform over sample realizations. Together with the teacher construction this proves the cold and fixed-gap parts of Theorem 1.1.

9 The critical endpoint

The same activation satisfies the hypotheses of the critical target-mass and canonical-state results proved in Appendix B. Here we check the activation-specific strict potential explicitly. For τ sufficiently small, (6) gives

$$|R(u)| \leq \varepsilon \min(|u|^3/3, 1).$$

Both ϕ and tanh are bounded by one and by $|u|$, so $|\phi(u)^2 - \tanh(u)^2| \leq 2\varepsilon \min(u^4, 1)$. For standard Gaussian Z and $0 \leq q \leq 2$,

$$\left|\mathbb{E}e^{\phi(\sqrt{q}Z)^2} - \mathbb{E}e^{\tanh(\sqrt{q}Z)^2}\right| \leq 6e\varepsilon q^2.$$

There is $c_0 > 0$ with $\log \mathbb{E}e^{\tanh(\sqrt{q}Z)^2} \leq q - c_0 q^2$ on $[0,2]$: for $G(u) = e^{\tanh^2 u}$, direct differentiation gives $G'' < 2G$ away from zero, and the heat equation gives strict inequality at positive q; Taylor expansion at zero gives $q - q^2 + O(q^3)$. Compactness supplies c_0. Taking $\varepsilon \leq c_0/(12e)$ proves

$$\log \mathbb{E}e^{\phi(\sqrt{q}Z)^2} \leq q - (c_0/2)q^2 \qquad (0 \leq q \leq 2). \tag{20}$$

We may impose this further smallness on τ without changing any preceding argument. The global derivatives through order five are bounded by (5), and the weighted second-derivative estimate at zero gives $\phi'''(0) = -2 + O(\varepsilon) < 0$. That estimate also gives $|\phi''(u)| \leq 3\phi(u)$ for $u > 0$, after decreasing τ: integrate $|R'| \leq \varepsilon \operatorname{sech}^2$ to get $\phi(u) \geq (1-\varepsilon)\tanh u$ and use $|\phi''| \leq (1+\varepsilon)|\tanh''|$. All hypotheses in Appendix B are therefore met.

For the sample memorizer $g_S = 1 - 2\sum_{p \text{ negative in } S} \mathbf{1}_{\{p\}}$, Fourier variation is at most $1 + 2n$. The appendix proves that whenever $L \geq C\log(d+n+L)$, at the actual scale $\sigma^2 = 1/N$ and every $N \geq 2^{Cd}(d+n+L)^C$, the posterior gives mass at least $31/32$ to functions with Boolean circuits of size $\mathrm{poly}(d+n+L)$, uniformly over every sample of at most n inputs. Reduce $c > 0$ in Theorem 1.1 so that $n \leq e^{cd}$ and $L_d = Dd + K$ imply both the depth condition and this circuit-size bound being at most $2^{d/4}$. The definition of f_d then gives error at least $7/16$ for each such function. The same Gibbs calculation yields $e_{S,\sigma}(f_d) \geq 139/1024$. Finally $2^{Cd}(d+n+L_d)^C \leq e^{C'd}$ in this range. This proves the critical part and completes the theorem.

10 Why the activation cannot be efficiently evaluated

Proposition 10.1. *There are fixed constants $C, c' > 0$ such that approximating this ϕ to error 2^{-Cd} on all dyadic inputs of magnitude at most Cd and bit length at most Cd cannot be done by Boolean circuits of size less than $\exp(c'd)$ for all sufficiently large d. In particular no algorithm performs these evaluations in time polynomial in the input length and requested precision.*

Proof. Round the teacher's rational parameters to dyadics with $O(d)$ bits. Its one binomial-combination layer has a fixed norm, and its remaining amplifying layers have scalar weight two. Layerwise Lipschitz propagation therefore bounds the output error from parameter, arithmetic, and

activation-evaluation errors by $C_1 d\, 2^{Dd}\delta$ when every primitive error is at most δ. Taking $\delta = 2^{-C_2 d}$ with fixed sufficiently large C_2 preserves the global output signs. All preactivation arguments have magnitude $O(d)$ and dyadic bit length $O(d)$. There are polynomially many arithmetic operations and activation calls.

If one accurate activation evaluator had circuit size $\exp(c'd)$, replacing all activation calls by copies of it would give a circuit for f_d of size at most $\mathrm{poly}(d)\exp(c'd)$. Choosing fixed $c' < (\log 2)/4$ makes this smaller than $2^{d/4}$ eventually, contrary to the defining circuit hardness of f_d. Enlarge C to cover the required argument and precision constants. Polynomial running time would in particular give polynomial-size circuits, so is excluded as well. □

11 Scope

The activation is fixed for the entire dimension sequence and is computable by finite search and convergent analytic series. It is intentionally computationally contrived. The proof asserts no efficient evaluator for it: inverse-exponential accuracy at its lookup bands reveals circuit-hard truth tables. The teacher architecture, coefficient sizes, and coefficient precision remain polynomial in its depth.

There is no limiting inference to the exact tanh activation. Decreasing the perturbation size decreases the encoded signal and increases the fixed number of amplification layers. The decoder deliberately amplifies that small activation difference to a constant output margin; its constants are not uniform as the perturbation tends to zero.

Thus oddness, analyticity, monotonicity, concavity, exact slope one at zero, and arbitrary closeness to tanh in any fixed finite number of derivatives do not by themselves give a universal polynomial-depth learning theorem. The exact tanh activation is unchanged by this statement. No lower bound for the exact tanh, ReLU, or GELU activation follows from this construction.

A Universal finite-circuit capacity

We prove Proposition 2.1, including its parallel-depth assertion. All activations in this appendix may differ by neuron.

Relevant coordinates. Let T be the Frobenius product after the first matrix. If coordinate i is relevant, a bichromatic cube edge has output difference at least two, so $T\left\|W^1 e_i\right\| \geq 1$. Summing the squares shows that the number k of relevant coordinates is at most P^2. Freeze the others to signs without increasing the first Frobenius norm: the expected squared norm of the new bias column under independent random signs is the sum of the old bias and frozen-column squared norms. Some assignment attains at most this expectation. Put $R = \sqrt{k+1}$ and $B = 4 + \lceil P \rceil$. Also $P \geq 1$, since zero preservation and the margin imply

$$1 \leq \mathbb{E} z(x)^2 \leq T^2 \mathbb{E} \left\| W^1(1,x) \right\|^2 = T^2 \left\| W^1 \right\|_F^2 \leq P^2.$$

In particular $R \leq B$.

A short prefix can be compressed. For adjacent matrices W, V, let w_i be row i of W and v_i column i of V, with $a_i = \|w_i\|$, $b_i = \|v_i\|$, and $A_0 = \sum_i a_i b_i \leq \|W\|_F \|V\|_F$. Delete zero contributions. Normalize the incoming rows to $u_i = w_i/a_i$ and activations to $\psi_i(s) = \phi_i(a_i s)/a_i$. Sample m indices with probabilities $p_i = a_i b_i / A_0$. Each sampled neuron has row u_i, activation ψ_i,

and outgoing column $A_0 v_i/(mb_i)$. These activations remain zero-preserving and 1-Lipschitz. For every draw the new adjacent Frobenius product is exactly A_0, and its expected contribution to the next preactivation is the old one.

Let S be the preceding product. The incoming state norm is at most SR, and an unaveraged sampled vector has norm at most $A_0 SR$. Hilbert-space bounded-difference concentration and a union bound over X_k give some draw with uniform preactivation error $O(A_0 SR\sqrt{\log(8|X_k|)/m})$. The suffix after V has Lipschitz constant at most $P/(S\|W\|_F \|V\|_F)$. Thus final output error is at most $O(PR\sqrt{\log(8|X_k|)/m})$. For $h = O(\log B)$ choose $m = C(PRh)^2 \log(8|X_k|)$, and perform this replacement through the first $\min\{h, L-1\}$ hidden layers. The product never increases, and sufficiently large universal C gives total output error at most $1/4$. Double the readout: the resulting network has margin at least $3/2$, product $P_0 \le 2P$, and retained widths polynomial in B. A following unretained matrix may still have arbitrarily many rows, which does not affect these claims.

Two-ball refinement. The image of a radius-r ball under a matrix W is covered by at most two balls of radius $r\|W\|_F/\sqrt{2}$. Normalize $\|W\|_F = 1$ and let a be its leading singular value. If $a^2 \le 1/2$, one ball at zero works. Otherwise put $b^2 = 1 - a^2$, $c = (a^2 - b^2)/(2a)$, and use centers $\pm cu$ along the leading left-singular direction. For a unit input with leading right-singular coordinate t, the squared distance to the matching center is at most

$$(a|t| - c)^2 + b^2(1 - t^2).$$

It is convex in $|t| \in [0, 1]$, and both endpoint values are $1/(4a^2) \le 1/2$. Scaling and translation prove the claim. Coordinatewise Lipschitz activations preserve the covering radii.

Let S_j be the product of the first j modified matrix norms. Iteration gives at most 2^j balls of radius $RS_j 2^{-j/2}$. Replace each nonempty ball center by an actual hidden image, doubling its radius. Choose $h = \lceil 2\log_2(64B^2)\rceil$. If at least h hidden layers exist, normalize their final state by S_h. Every input is within $1/(8\overline{P})$ of an actual representative, where $\overline{P} = 4B$. Opposite-label representatives are at distance at least $2/\overline{P}$: the normalized suffix is P_0-Lipschitz and both outputs have margin at least one. There are at most $2^h = \mathrm{poly}(B)$ centers. If fewer hidden layers exist, retain the entire compressed network and its readout instead.

Normalize before rounding. Put $M_s = \|W^s\|_F$, $S_s = \prod_{i \le s} M_i$, $A_s = W^s/M_s$, and $\Psi_{s,i}(u) = \phi_{s,i}(S_s u)/S_s$. No M_s vanishes, since that would force all outputs to vanish. The normalized states satisfy

$$u_s = \Psi_s(A_s u_{s-1}), \qquad \|A_s\|_F = 1, \qquad \|u_s\| \le R.$$

Every $\Psi_{s,i}$ is zero-preserving and 1-Lipschitz. Choose inverse-polynomial meshes in B for the matrix entries and for scalar tables of $\Psi_{s,i}$ on $[-R-2, R+2]$. If matrix Frobenius error is at most η and scalar lookup error at most 2δ, the state error obeys

$$e_s \le (1 + \eta)e_{s-1} + \eta R + 2\sqrt{C}\,\delta,$$

where $C = \mathrm{poly}(B)$ bounds the retained widths. Taking η, δ below suitable inverse polynomials in B, C, h gives $e_s < 1/(256\overline{P})$ throughout. Approximate affine inputs remain in the stored interval. Store the normalized representatives to the same Euclidean accuracy. Squared-distance comparisons at threshold $1/\overline{P}^2$ distinguish a close correct-label center from every opposite-label center. The output is positive precisely when a positive-labeled center lies below that threshold. In the short-network case, round the normalized readout as well; its margin at least $1/\overline{P}$ dominates these errors.

All tables have polynomial length in B, with $O(\log B)$ bits per entry. Fixed-precision sums, products, comparisons, and lookups therefore give a Boolean circuit of size $\mathrm{poly}(B)$ with universal degree. Carry-save addition trees followed by carry propagation take $O(\log B)$ depth per retained layer, giving total $O(\log^2 B)$ Boolean depth. Alternatively each fixed affine comparison of preceding state bits is a threshold gate. Interval tests and table-bit selection take constant threshold depth per layer. Squaring the final $O(\log B)$-bit differences uses polynomial-size lookup tables, and threshold gates sum the squares. The threshold depth is $O(\log B)$. This proves Proposition 2.1.

B Critical target mass and posterior circuits

This appendix proves the finite-width critical input used in the main theorem. Throughout, constants may depend on one fixed activation ϕ. We assume that ϕ is odd, strictly increasing, bounded by one, and 1-Lipschitz; that it is C^5 with bounded derivatives through order five, concave on $(0,\infty)$, and satisfies

$$\phi'(0) = 1, \qquad \phi'''(0) < 0, \qquad \phi'(u) \geq c_0 \operatorname{sech}^2 u.$$

Writing $\beta = \lim_{u\to\infty} \phi(u) > 0$, assume $\beta - \phi(u) \leq C(1+u)^{-a}$ for $u \geq 0$ and some $a > 0$. The final assumption is the strict potential

$$\log \mathbb{E} e^{\phi(\sqrt{q}Z)^2} \leq q - cq^2, \qquad 0 \leq q \leq 2, \tag{21}$$

where Z is standard Gaussian. The main text verifies all these conditions for the activation constructed there.

Set $H = L - 1 \geq 1$. For a finite input set X, a controlled critical path starts at $Q_0(x,y) = (1 + x \cdot y)/d$ and consists of

$$\rho_j \ll \mathcal{N}(0, Q_{j-1}), \quad D_j = \mathrm{KL}(\rho_j \| \mathcal{N}(0, Q_{j-1})), \quad Q_j(x,y) = \mathbb{E}_{\rho_j} \phi(G_x)\phi(G_y).$$

Singular references are permitted. Its output $z \in \operatorname{range} Q_H$ has readout cost $r = z^{\mathsf{T}} Q_H^{\dagger} z/2$, and total action $A = \sum_j D_j + r$. Let $J_L(f)$ be the infimum of this action over full-cube paths with $f(x)z(x) \geq 1/2$ everywhere. These definitions will be related to actual finite-width probabilities below.

B.1 Scalar control, cooling, and a target program

For $q > 0$ let $V(q)$ be the infimum, over finite scalar paths starting at $q_0 = q$, of

$$\sum_{j=1}^{m} \mathrm{KL}(\eta_j \| \mathcal{N}(0, q_{j-1})) + \frac{1}{8q_m}, \qquad q_j = \mathbb{E}_{\eta_j} \phi(G)^2.$$

The length-zero path is included.

Lemma B.1 (Scalar baseline and short paths)**.** *The function V is nonincreasing and has a finite positive limit $V_* := \lim_{q\downarrow 0} V(q)$. For a fixed C,*

$$0 \leq V_* - V(q) \leq Cq \quad (0 < q \leq 2).$$

A scalar transition of entropy D from variance q to q' satisfies

$$D \geq V(q) - V(q'), \qquad D \geq q' - q + cq^2 \quad (q \leq 2).$$

For every sufficiently small $p > 0$ there is a strictly increasing scalar path starting at p, of total cost at most $V(p) + p$, whose length is at most C/p and for which

$$\sum_j q_j \le C \log(2/p), \qquad \sum_j \sqrt{D_j} \le C \log(2/p). \tag{22}$$

Proof. Rescaling a controlled preactivation when its reference variance increases preserves its Gaussian relative entropy and increases its ϕ-square moment. Applying this successively proves monotonicity. Oddness, $\phi'(0) = 1$, and bounded third derivatives give $F(u) := \mathbb{E}\phi(\sqrt{u}Z)^2 \ge u - Cu^2$ for small u; outside a fixed neighborhood of zero use Gaussian tails and $|\phi(z)| \le |z|$. Inflating the reference variance by $1 + Kq$, with fixed large K, therefore increases a small q by between $c_1 q^2$ and $C_1 q^2$ at Gaussian cost at most $C_2 q^2$. Continuous adjustment of the inflation hits any prescribed small endpoint p exactly. Telescoping the cost-to-increment ratio gives a bridge from arbitrarily small positive variance to p costing at most Cp. A direct readout at one fixed p proves finiteness of V_*, and appending a nearly optimal suffix proves $V_* - V(p) \le Cp$. Boundedness extends this inequality to $q \le 2$. For starting $q \le 1$, every final variance, including that of a length-zero path, is at most one; hence $V_* \ge 1/8$. The Bellman inequality follows by appending a nearly optimal suffix, and entropy duality applied to (21) gives dissipation.

Delete each nonincreasing transition of a near-optimal path and rescale the subsequent controls as above. Cost does not increase, so the path may be taken strictly increasing. Its cost is bounded by a fixed constant B. Gaussian entropy duality and $\phi(z)^2 \le z^2$ bound each ratio q'/q by a constant depending on B. The entropy spent before a state q is at most $p + Cq$, by the scalar suffix bound. In a dyadic band $u \le q < 2u$, its expenditure, including its exit transition, is therefore $O(u)$: the exit state is at most a fixed multiple of u. Summed dissipation bounds its length by $O(1/u)$, apart from at most one exit transition. The band contributes $O(1)$ to both sums in (22), using Cauchy–Schwarz for the second. Summing the geometric length bounds and logarithmically many bands proves the assertion. States above a fixed small constant occupy only a bounded number of transitions by the same dissipation. □

Lemma B.2 (An exact low-cost Fourier seed)**.** *Let $A_F(f) = \sum_T |\widehat{f}(T)|$. There is a first-row controlled law, with auxiliaries and a score b_1, such that*

$$\mathbb{E}b_1^2 = 1, \quad \mathbb{E}[b_1 \phi(w \cdot (1, x))] = \sqrt{v_1} f(x), \quad \mathbb{E}\phi(w \cdot (1, x))^2 = q_1$$

with q_1 independent of x, relative entropy at most $1/H$, and $v_1^{-1}, \|b_1\|_\infty$ bounded by a fixed polynomial in $(d + H + 2)A_F(f)$. The reference weight marginal is exactly $\mathcal{N}(0, I/d)$.

Proof. For centered Gaussian G of variance at most two, split at $G = -M/2$ to obtain

$$\beta - \mathbb{E}\phi(M + G) \le CM^{-a} + Ce^{-M^2/16}.$$

A parity on k coordinates is a signed sum of k threshold steps in their Hamming sum and a constant, with coefficient magnitudes at most one. Each step has an affine argument separated from zero by one. Shift first-row Gaussian means by a polynomially large multiple of these affine coefficients, and mix the signed detectors; reflection of the row implements a negative sign. Their normalized mixture has mean g with $\chi_T(x)g(x) \ge c/(k + 1)$ everywhere. Every component costs at most $D_* = \mathrm{poly}(d)$ in Gaussian KL.

Average this law over cube translations $x \mapsto sx$, reflecting the output row by $\chi_T(s)$. Its mean becomes exactly $c_T \chi_T(x)$, where $c_T = \mathbb{E}_x \chi_T(x)g(x) \ge c/(d + 1)$. Its second-moment diagonal is constant. The transformations preserve the Gaussian reference, and entropy convexity preserves

the cost D_*. Mix these laws, with reflections according to sign $\widehat{f}(T)$, in proportions $|\widehat{f}(T)|/c_T$. The resulting law has mean f/Z_f, constant diagonal, cost at most D_*, and $Z_f \le C(d+1)A_F(f)$.

Dilute this law into the free first-row law with probability $\theta = (HD_*)^{-1}$, increasing $D_* \ge 1$ if needed. Retain its Bernoulli tag B_0 and use $b_1 = B_0/\sqrt{\theta}$. The coefficient is $\sqrt{\theta}f/Z_f$, so $v_1 = \theta/Z_f^2$. In the reference use the same tags and free Gaussian weights in every branch. Its weight marginal is the original Gaussian, and the joint entropy is at most $\theta D_* \le 1/H$. All claimed bounds follow. In particular, no entropy proportional to the number of translation or Fourier tags has been charged. □

Lemma B.3 (A scalar-channel target program)**.** *Put $R = (d+L+2)A_F(f)$ and $S = 1+\log R$. If $H \ge CS$, there is an exact H-layer controlled program with scores b_j and deterministic positive components v_j, residual diagonals $r_j \ge 0$, such that*

$$\mathbb{E}b_j^2 = 1, \quad \mathbb{E}[b_j h_j(x)] = \sqrt{v_j} f(x), \quad \mathbb{E}h_j(x)^2 = v_j + r_j \le 1.$$

Its charged scalar costs D_j obey

$$A := \sum_j D_j + \frac{1}{8v_H} \le V_* + CS^2/H, \quad K := \sum_{j=2}^{H} \sqrt{D_j} \le CS, \quad |b_j| \le B \le e^{CS}. \tag{23}$$

For $j \ge 2$, the controlled row uses a scalar Z, of entropy cost at most D_j relative to a reference with standard Gaussian Z marginal, and independent Gaussian residual noise. At one input its feature is $\phi(\sqrt{v_{j-1}}f(x)Z + \sqrt{r_{j-1}}G)$. The score depends on Z and auxiliaries, not on G.

Proof. We supply the scalar estimates and the depth accounting. Write $b(u) = \mathbb{E}\phi'(\sqrt{u}Z)$. Taylor expansion, with bounded fifth derivatives and Gaussian tails, gives

$$F(u) = u + \phi'''(0)u^2 + O(u^3), \quad b(u) = 1 + \tfrac{1}{2}\phi'''(0)u + O(u^2), \quad \frac{ub(u)^2}{F(u)} = 1 + O(u^2). \tag{24}$$

Also $F(u) \le u - cu^2$ by (21). In a free layer, Gaussian integration by parts gives at least $vb(q)^2$ as the component generated from vff^{T} at diagonal q. The component-to-diagonal ratio therefore loses factors $qb(q)^2/F(q)$. Their logarithms are $O(q^2)$ near zero, and they have a positive lower bound on each remaining compact interval. Since free dissipation bounds $\sum q_j^2$, cooling to the first diagonal at most δ takes $O(1/\delta)$ layers and retains component at least $c\delta v_1$. If already below δ, skip cooling; the same lower bound holds. Cooling costs zero entropy.

The estimate

$$1 - \phi'(z) \le C\phi(z)^2 \tag{25}$$

follows near zero by odd C^3 regularity and away from zero by positivity of ϕ. Inflate only the target Gaussian coordinate by variance factor $1+e$. Its cost is $[e - \log(1+e)]/2 \le e^2/4$. If the current component and residual diagonal are v, r and $q = v + r$, the full conditional-mean component and new residual satisfy

$$v' \ge (1+e)vb(q+ev)^2 \ge (1+e)v[1 - C(q+ev)], \qquad r' \le r \tag{26}$$

in the small-variance range. The first inequality is the first Hermite projection; the last follows from conditional Gaussian Poincare. The component is continuous and increasing in e, and at $e = 0$ it is at most v. Thus at $q \le \delta$ one can hold v exactly with $e = O(\delta)$, at cost $O(\delta^2)$, preserving $q \le \delta$. If $r \le \delta$ and $v \le p$, one can prescribe

$$v' = \min\{p, v[1 + c_1(\delta + v)]\}$$

with $e = O(\delta + v)$ and cost $O((\delta + v)^2)$. From $v_0 = \delta e^{-C_1 S}$ this takes $O(S/\delta)$ steps and costs $O(p + S\delta)$. Indeed, on full steps, $(\delta + v)^2 = O(\Delta v + \delta \Delta v/v)$; their square-root costs sum to $O(\log(p/v_0) + 1) = O(S)$ when p/δ is fixed. The prescribed sequence fixes the length independently of the residual and any preceding holding time.

At $v = p$, $r \le \delta \le p$, hold the full component exactly. The total Gaussian preactivation variance lies in $[p, Cp]$, and

$$\mathbb{E}\phi'(\sqrt{u}Z)^2 = 1 + \phi'''(0)u + O(u^2) \le 1 - cp.$$

The derivative form of conditional Poincare now gives $r' \le (1 - cp)r$. Thus $O(\log(H + 2)/p)$ purification stages leave $r \le \delta H^{-4}$, at cost $O(p \log(H + 2))$ and square-root cost $O(\log(H + 2))$.

For the scalar suffix, $|\phi''(z)| \le C\phi(z)$ for $z > 0$: use oddness and C^3 regularity near zero, positivity on compact intervals, and bounded ϕ'' elsewhere. If P_r denotes Gaussian heat convolution, then

$$e^{-Cr/2}\phi(z) \le P_r\phi(z) \le \phi(z), \qquad z \ge 0. \tag{27}$$

Indeed $g = \phi'' + C\phi$ is odd and nonnegative on the positive half-line. Pairing the Gaussian density at $y, -y$ shows $P_s g(z) \ge 0$ for $z \ge 0$, so $\partial_s[e^{Cs/2}P_s\phi(z)] \ge 0$. The upper bound follows in the same way from concavity and the odd function ϕ''. Concavity also gives $\phi(\alpha z) \ge \alpha\phi(z)$ for $0 \le \alpha \le 1$ and $z \ge 0$.

Lift the short scalar path of Lemma B.1 starting at p. For a nominal law η from variance q and an actual component $v \le q$, use physical scalar $\sqrt{v/q}X$, $X \sim \eta$. Its Gaussian relative entropy is unchanged. Equation (27) and concavity imply

$$e^{-Cr}(v/q) \le v'/q' \le 1,$$

and the residual remains at most r. Starting at $v = q = p$, the lift of $m \le C/p$ stages therefore retains at least $e^{-Cm\delta H^{-4}}$ of every nominal component. Since $\delta \le p$, all components stay at least $p/2$, and the extra readout cost is $O(H^{-4})$. Its charged square-root cost is $O(\log(2/p))$ by (22).

Choose $\delta = B_1 S/H$ and $p = B_2 S/H$, with fixed sufficiently large $B_2 \ge B_1$, and then increase the constant in $H \ge CS$ so both scales are small. After cooling reduce the component to $v_0 = \delta e^{-C_1 S}$, which is permitted by Lemma B.2. Cooling, prescribed growth, purification, and the chosen scalar suffix occupy at most $H - 1$ stages after choosing B_1, B_2. Insert exactly the unused number of holding stages before growth. The total cost is

$$A \le H^{-1} + V(p) + p + C[H\delta^2 + p + S\delta + p\log(H + 2) + H^{-4}] \le V_* + CS^2/H.$$

Holding contributes $O(H\delta) = O(S)$ to the square-root sum; the other contributions were bounded above, proving $K \le CS$. This is a bound on the charged costs themselves, not an inference from smaller marginal entropies.

The full conditional-mean score is $P_r\phi(\sqrt{v}Z)/\sqrt{v'}$, of unit second moment. To retain a smaller prescribed component $0 < v'' \le v'$, replace it by

$$\sqrt{v''/v'}\, b + \sqrt{1 - v''/v'}\, U,$$

where U is an independent random sign with the same reference law. This has unit second moment, the desired coefficient, and absolute bound $1 + (v'')^{-1/2}$, with no extra entropy. Every retained component in the construction is at least e^{-CS}, proving the score bound. At each stage an independent Gaussian residual vector with the current residual covariance realizes the stated full-cube law. Only its constant diagonal enters the one-input identities. □

B.2 Finite-width realization without a rate-limit argument

Lemma B.4 (Finite scalar-channel probability)**.** *Suppose a program as in Lemma B.3 has action A, distortion $K = \sum_{j\geq 2}\sqrt{D_j}$, and score bound $B \geq 1$. For every $\tau > 0$ and every integer*

$$N \geq CB^4H^4e^{4K}(A+1)^6(1+\tau^{-1})^4[\log(2|X|)+\log(H+A+\tau^{-1}+3)], \tag{28}$$

the original critical Gaussian prior satisfies

$$\mu_N\{f(x)z_W(x) > 1/2 \ (x \in X)\} \geq e^{-N(A+\tau)}.$$

The lemma uses only boundedness and the unit Lipschitz constant of the activation, in addition to the specified program.

Proof. For N independent score/feature rows set

$$M = N^{-1}\sum_i b_i^2, \quad \widehat{c}(x) = \frac{N^{-1}\sum_i b_i h_i(x)}{\sqrt{M}}, \quad \widehat{q}(x) = N^{-1}\sum_i h_i(x)^2, \quad \widehat{s}(x) = \sqrt{\widehat{q}(x)-\widehat{c}(x)^2}.$$

If the three raw moments have errors at most $\alpha \leq 1/4$, then, writing $c = \mathbb{E}bh$ and $s = \sqrt{\mathbb{E}h^2 - c^2}$,

$$\max_x \sqrt{(\widehat{c}-c)^2 + (\widehat{s}-s)^2} \leq 8\sqrt{\alpha}. \tag{29}$$

For example $|\widehat{c}-c| \leq 4\alpha$ and the squared residual error before taking square roots is at most 13α. The coefficients $a_i = b_i/(N\sqrt{M})$ have norm squared $1/N$; the empirical kernel decomposes exactly as $K_{\text{emp}} = \widehat{c}\widehat{c}^{\mathsf{T}} + R_{\text{emp}}$ with $R_{\text{emp}} \succeq 0$.

Condition on the entire preceding construction and set $u = \sqrt{N}a$. For a fresh row draw its prescribed scalar (Z,U) and independent standard Gaussian $\xi \in \mathbb{R}^N$, and use

$$w = [Zu + (I - uu^{\mathsf{T}})\xi]/\sqrt{N}. \tag{30}$$

Under the reference, Z is standard Gaussian, so the incoming weight marginal is exactly $\mathcal{N}(0, I/N)$. Its controlled relative entropy is at most D_j by data processing. At one input the preactivation has law $\widehat{c}(x)Z + \widehat{s}(x)G$, with independent standard G. Couple this to the ideal preactivation using the same Z, U, G. Gaussian entropy duality gives $\mathbb{E}Z^2 \leq e^{2\sqrt{D_j}}$, so the preactivation and feature RMS errors are at most $e^{\sqrt{D_j}}$ times the preceding metric error. Orthogonal projection onto the common unit score, followed by the reverse triangle inequality for residual norms, gives the same bound for the population coefficient/residual metric. Fresh sampling and (29) consequently give

$$e_H \leq 8He^K\sqrt{\alpha}. \tag{31}$$

Only one-input comparison couplings are used; the sampled network keeps its usual joint residual Gaussian law on all inputs.

The observables $b^2, bh(x), h(x)^2$ are bounded by $B^2, B, 1$. Conditional Hoeffding, a union bound over $2|X|+1$ observables, and successive conditioning over layers give all moment estimates with probability at least $1-\kappa/2$ if

$$N \geq CB^4\alpha^{-2}\log[8H(2|X|+1)/\kappa].$$

Stop controlling if $M \notin [1/2, 3/2]$ and thereafter use the prior. This defines the law on all histories, while the good event never stops. Choose

$$\kappa = \min\{1/4, \tau/[32(A+1)]\}, \qquad \alpha = \left[\frac{\kappa\sqrt{v_H}}{64He^K}\right]^2.$$

Then the final coefficient has signed margin at least $\sqrt{v_H}(1-\kappa/8)$. On this event shift the readout mean to ta, where $t=(1+\kappa)/(2\sqrt{v_H})$; otherwise use zero mean. Its mean output is at least $1/2+\kappa/4$ in the correct direction. Its noise variance is at most $1/N$ at each input. If $N \geq C\kappa^{-2}\log(4|X|/\kappa)$, the maximum noise is less than $\kappa/8$ with probability at least $1-\kappa/2$. Thus the controlled law Q_N gives the strict target event probability at least $1-\kappa$.

Use the same reference auxiliaries as in the program. The reference weight marginal in (30) remains Gaussian even under the history-dependent orthogonal decomposition. Conditional chain rule and data processing give hidden-block cost at most ND_j; stopping only reduces this bound. The readout shift costs $Nt^2/2$. After discarding auxiliaries,

$$\mathrm{KL}(Q_N\|\mu_N) \leq N\left[\sum_j D_j + \frac{(1+\kappa)^2}{8v_H}\right] \leq N(A+3A\kappa).$$

Binary relative entropy for the target event implies

$$-N^{-1}\log\mu_N(E_f) \leq \frac{A+3A\kappa+N^{-1}\log 2}{1-\kappa} \leq A+\tau$$

when $N \geq 8/\tau$. Since $v_H^{-1} \leq 8A$, the width bound (28) implies all the numerical conditions above. □

Theorem B.5 (Critical raw target mass)**.** *For fixed ϕ satisfying the hypotheses of this appendix, there are constants C_0, C_1, C_2 such that, if $R=(d+L+2)A_F(f)$ and $L \geq C_0\log R$, then at every integer $N \geq R^{C_1}$,*

$$\mu_N\{f(x)z_W(x) > 1/2 \ (x \in X_d)\} \geq \exp\left[-N\left(V_* + C_2\frac{\log^2 R}{L}\right)\right].$$

Proof. Apply Lemma B.4 to Lemma B.3 with $\tau = 1/H$. Each factor in (28) is at most e^{CS}: $B \leq e^{CS}$, $K \leq CS$, $H, d \leq e^S$, and $A+1 \leq C(1+S)$. Also $\log|X_d| = d\log 2$. The additional $1/H$ action is absorbed into CS^2/H. Enlarging constants handles $H = L-1$ and $S = 1+\log R$. □

B.3 Near-baseline paths have parallel circuits

For $D \geq 0$ define $G(D) \geq 1$ by $G(D) - 1 - \log G(D) = 2D$. Gaussian entropy duality implies

$$\mathbb{E}_\rho\langle g,u\rangle^2 \leq G(D)\,\|u\|^2, \qquad \sqrt{G(D)} \leq e^{\sqrt{D}}, \qquad G(D) \leq 4D+2 \tag{32}$$

for any tilt of a standard Gaussian with entropy at most D. Indeed projection onto $\langle g,u\rangle/\|u\|$ gives $D \geq (m-1-\log m)/2$ when its second moment $m > 1$; writing $m = e^{2v}$ gives $(m-1-\log m)/2 \geq v^2$.

Lemma B.6 (Variance and entropy budgets)**.** *For a full-cube globally $1/2$-margined path of action $A \leq V_* + \epsilon$, $\epsilon \geq 0$, put $q_j = \max_x Q_j(x,x)$. Then*

$$\sum_{j=0}^{H-1} q_j + \sum_{j=1}^{H}\sqrt{D_j} \leq C[1+\log(H+2)+\sqrt{H\epsilon}].$$

Proof. All one-input variances are positive, since a zero variance would propagate at finite entropy and force zero final output. Decrease c so $q - cq^2$ is increasing on $[0,2]$. Projecting onto an input attaining the new maximum and using Lemma B.1 yields

$$D_j \geq q_j - q_{j-1} + cq_{j-1}^2.$$

The scalar suffix at an input attaining q_i, including its readout, costs at least $V(q_i)$. Thus

$$\sum_{j\le i} D_j \le \epsilon + Cq_i. \tag{33}$$

These inequalities concern full layer entropies. No two-input contraction property of the activation is needed.

For an interval of t transitions whose pre-states and endpoint are at least u, dividing dissipation by the squared pre-state and using $(q'-q)/q^2 \ge 1/q - 1/q'$ gives $t \le C(E/u^2 + 1/u)$, where E is its entropy expenditure. An initial prefix of length t and expenditure E also obeys

$$\sum_{j<t} q_j \le C\log(t+2) + C\sqrt{tE} \tag{34}$$

when total entropy is bounded by a fixed constant. To prove this, write $F_0(v) = v - cv^2$, compare q_j to $r_j + s_j$ with $r_{j+1} = F_0(r_j)$, $s_{j+1} = F_0(s_j) + D_{j+1}$, $r_0 = q_0 \le 2$, $s_0 = 0$, and decrease c so F_0 is increasing on the resulting bounded range. Since $F_0(r+s) \le F_0(r) + F_0(s)$, induction proves the comparison. The free sum is $O(\log(t+2))$, and $c\sum s_j^2 \le E$, proving (34).

If ϵ exceeds a fixed positive constant, summed dissipation and Cauchy–Schwarz prove the claim. Otherwise total entropy is bounded. Set $a = \max\{\epsilon, (H+1)^{-1}\}$, $b = \min_j q_j$, and $u_0 = \max\{a, b\}$. If u_0 exceeds a fixed small constant, the occupation bound gives bounded H. Otherwise use dyadic levels $u_j = 2^j u_0$ until a fixed constant, and their last-exit indices $\tau_j = \max\{i : q_i \le u_j\}$. There are $O(\log(H+2))$ bands. The initial prefix has entropy $O(\epsilon + 1/H)$ if $b \le a$; then (34) and Cauchy–Schwarz bound its two contributions as claimed. If $b > a$, its entropy is $O(b)$ and its length $O(1/b)$, giving state sum $O(\log(H+2))$ and square-root entropy $O(1)$. A later nonempty last-exit block has entropy $O(u_j)$ by (33); after removing its first transition its pre-states and endpoint exceed u_j, so its length is $O(1/u_j)$. Every state in it is $O(u_j)$ because its starting state is at most u_j and any increase is at most spent entropy. Both contributions are $O(1)$. The final tail has bounded length and contributions. Summing proves the assertion. □

Theorem B.7 (Critical canonical circuit theorem)**.** *A globally 1/2-margined path of action $A \le V_* + \epsilon$ gives a nonuniform threshold circuit of size $R_1^{O(1)}$ and depth $O(\log R_1)$, and a bounded-fan-in Boolean circuit of size $R_1^{O(1)}$ and depth $O(\log^2 R_1)$, where*

$$T = 1 + \log(H+2) + \sqrt{H\epsilon}, \qquad R_1 = (d + H + A + 2)e^{CT}.$$

All constants depend only on the fixed activation.

Proof. We give the anchor reduction, finite-dimensional state, evaluator, and parallel iteration explicitly.

Anchor reduction. Let $I_j(U)$ be the entropy of the layer-j preactivation marginal on an anchor set U. It increases with U and is at most D_j. Choose $\rho = [CH(A+1)]^{-1} \le 1/64$ and a cutoff $\delta > 0$. Greedily add a query whenever $I_j(U \cup \{x\}) - I_j(U) > \delta$ for some layer, or projection of the final readout onto the anchor feature span changes its output there by more than ρ. There are at most A/δ entropy additions and $2r/\rho^2$ readout additions: each latter increases the projection's squared norm by at least ρ^2, since final feature norms are at most one.

Replace each row density by its conditional expectation $w_j(G_U)$ given the anchors. It has entropy $I_j(U) \le D_j$ and preserves the original anchor marginal exactly. The conditional query KL increment is at most δ. Conditional maximal coupling and Pinsker give expected squared feature error at most

$4\sqrt{\delta/2}$, since features are bounded by one. For kernels sharing their anchor block, compare a query by its projection onto the common anchor span and its orthogonal residual norm. This defines a metric d_U equal to the RMS error in a Gaussian coupling sharing the anchors and one residual standard Gaussian. The preceding replacement has d_U error $O(\delta^{1/4})$. For a later Gaussian reference with the same anchor block, tilt this one-query coupling by the common w_j. Its entropy is $I_j(U)$; (32) and the Lipschitz property multiply the input error by at most $e^{\sqrt{D_j}}$. Thus the complete error is at most $CH\delta^{1/4}\exp(\sum_j \sqrt{D_j})$. Choose δ so this is at most $\rho/\sqrt{\max\{2r,1\}}$. Lemma B.6 makes the anchor count polynomial in R_1.

These replacements define a *genuine* full-cube controlled path, with the exact original anchor blocks, density depending only on anchors, and no greater row entropies. The projected readout loses at most 2ρ of margin. Scaling it by $(1-4\rho)^{-1}$ restores margin $1/2$ at action increment $O(1/H)$. Its coefficient vector has norm $O(\sqrt{A+1})$. The genuine path therefore satisfies the budgets of Lemma B.6 with bound CT.

Canonical finite states. Initially $u_0(x) = (1,x)/\sqrt{d} \in \mathbb{R}^{d+1}$. Given states in $\mathbb{R}^K$, let $g \sim \mathcal{N}(0, I_K)$ tilted by $w_j((\langle g, u(a)\rangle)_{a\in U})$, and write this law as ρ_j. In $L^2(\rho_j)$ put $\Phi(u) = \phi(\langle g, u\rangle)$ and let V span all K coordinate functions g_i and all anchor features $\Phi(u(a))$. For an orthonormal basis $e_1, \dots, e_M$, $M \le K + |U|$, define

$$F(u) = (\langle e_b, \Phi(u)\rangle_{b=1}^{M}, \|(I - P_V)\Phi(u)\|). \tag{35}$$

Pythagoras preserves every diagonal. At anchors the last coordinate is zero, so every query-anchor cross moment and the entire anchor block are preserved exactly. Future anchor-plus-one-query Gaussian laws are determined by precisely these moments. Induction gives $d_U = 0$ relative to the genuine anchor-only path at every stage, and identical anchored readout. The canonical full covariance is not asserted to be a valid original transition. Its state dimension is at most $d + 1 + H(|U| + 1)$, polynomial in R_1.

Since the linear map $u \mapsto \langle g, u\rangle$ lies in V, write $F(u) = A_j u + B_j(u)$, with $\|A_j\|_{\mathrm{op}} \le \alpha_j := \sqrt{G(D_j)}$. The map taking a Hilbert residual to its V projection and its orthogonal norm is 1-Lipschitz. Moreover $|\phi'(s) - 1| \le C\min(s^2, 1)$ by odd C^3 regularity and $0 \le \phi' \le 1$. For $\|u\|, \|v\| \le \sqrt{q}$, put $X = g\cdot u$, $Y = g\cdot v$, $Z = g\cdot(u-v)/\|u-v\|$. The squared normalized residual difference is at most a fixed multiple of $W = Z^2\min\{1, \max(|X|,|Y|)^4\}$. For small fixed $c > 0$,

$$e^{cW} - 1 \le cZ^2(X^4 + Y^4)e^{cZ^2}, \qquad \log \mathbb{E}_{\mathrm{Gauss}} e^{cW} \le Cq^2.$$

For the second inequality condition X or Y on Z; its fourth moment is at most $Cq^2(1 + Z^4)$, regardless of correlations. Entropy duality consequently gives

$$\mathrm{Lip}(B_j) \le C(q + \sqrt{D_j}), \quad \sum_j \mathrm{Lip}(B_j) \le CT, \quad \prod_j \alpha_j \le e^{CT} \tag{36}$$

on the actual state balls. Here use the genuine path's diagonal budget and the original path's entropy budget; alternatively take D_j to be the actual anchor entropy. Extend each B_j by radial projection. Round the radius upwards on mesh $R_1^{-c_1}$ and impose that same lower floor, with fixed large c_1. This adds only a constant to the summed squared radii and preserves the actual path.

Polynomial finite evaluation. For any current law ρ, basis vector $e = (e_b)$, and polynomial input radius, write

$$a(u) = \mathbb{E}_\rho e(g)\phi(g\cdot u), \quad b(u) = \mathbb{E}_\rho \phi(g\cdot u)^2, \quad F(u) = (a(u), \sqrt{b(u) - \|a(u)\|^2}).$$

Equation (32) gives $\mathbb{E}_\rho \|g\|^2 \le KG(D)$. Truncate to $\|g\| \le M_0$. If the discarded probability is p, Bessel's inequality bounds the discarded vector expectation uniformly by $\sqrt{p}$, and the scalar expectation by p. A polynomial M_0 gives any inverse-polynomial accuracy. On the retained set sample with importance density $(1+\|e\|^2)/Z_0$ relative to ρ, normalized there, where $Z_0 \le M+1$. The vector and scalar integrands have norms at most $Z_0/2, Z_0$, and Lipschitz bounds at most $Z_0M_0/2, 2Z_0M_0$. An input net has logarithmic cardinality $O(K\log(2+\mathrm{poly}(R_1)/\theta))$. For bounded Hilbert-valued samples, expected mean error is at most the bound divided by the square root of the sample count; bounded differences adds an exponential tail. A union bound on this net therefore gives polynomially many nodes approximating a, b uniformly to error θ. Lipschitzness extends the estimate to the whole ball. The retained nodes and all importance-weighted coefficients have polynomial magnitude, irrespective of an anchor condition number. Since $\|a\| \le 1$, choosing $\theta = O(\eta^2)$ and clipping the approximate radicand at zero gives error $O(\eta)$ in F.

Round these constants, the matrix A_j, and all arithmetic to inverse-polynomial accuracy. Scalar ϕ values need only a polynomial mesh, since ϕ is 1-Lipschitz; no algorithm for evaluating ϕ is assumed. Every word has $O(\log R_1)$ bits. Unary and binary word operations have polynomial-size truth tables. A sum of polynomially many such words has polynomial integer range after a common fixed-point scaling; threshold tests against each possible integer and a bit decoder compute it in constant threshold depth. Signed integer weights of polynomial magnitude can be replaced by repeated inputs and complements. Squares, square roots, comparisons, clipping, and radial projection have the same bounds; division occurs only above the inverse-polynomial radius floor. Thus one canonical residual evaluator has polynomial size and constant threshold depth, uniformly on the required polynomial ball.

Parallel iteration. Set $S_j = \prod_{i\le j}\alpha_i$, $v_j = u_j/S_j$. The normalized recursion is $v_j = C_j v_{j-1} + H_j(v_{j-1})$, where $C_j = A_j/\alpha_j$ is a contraction and $\mathrm{Lip}(H_j) \le \ell_j := C(q_{j-1} + \sqrt{D_j})$ with $\Lambda := \sum_j \ell_j \le CT$. All S_j are polynomial in R_1. Precompute the interval products $C_{j:i} = C_j \cdots C_i$ as nonuniform constant matrices; their operator norms and entries are at most one. Starting with $v_j^{(0)} = C_{j:1}v_0$, compute all times simultaneously by

$$v_j^{(k+1)} = C_{j:1}v_0 + \sum_{i=1}^{j} C_{j:i+1}H_i(v_{i-1}^{(k)}).$$

Strict time ordering gives $\max_j \left\|v_j^{(k)} - v_j\right\| \le 2\sqrt{2}\Lambda^k/k!$: expansion uses distinct ordered indices, whose products sum to at most $\Lambda^k/k!$. Thus $O(\log R_1)$ iterations give any fixed inverse-polynomial accuracy. Uniform additive error δ_0 per update is amplified by at most $\sum_{k\ge0}\Lambda^k/k! = e^\Lambda = R_1^{O(1)}$. The clipped residuals have bounded ranges, so all exact and inexact iterates stay in a fixed polynomial ball. Rounded interval products, quadrature, and arithmetic contribute such additive errors; divide primitive tolerances by a polynomial to absorb the number of terms.

There are polynomially many coordinates and times, each update has constant threshold depth, and there are $O(\log R_1)$ updates. The final rescaling and readout amplify errors only polynomially; choose precision so the output error is below $1/16$. This preserves all signs. Replacing each polynomial-fan-in majority gate by a logarithmic-depth Boolean summation circuit proves the Boolean bound. □

B.4 A finite-event certificate and the full-cube ceiling

Lemma B.8 (Bounded-dual extraction)**.** *On m inputs put $k = m(m+1)/2$. For any raw event E of critical prior probability at least e^{-NA}, any $\tau > 0$, and $0 < \Delta \le 1$, there is a realization in E and*

laws relative to its empirical preceding Gaussian covariances such that

$$\max_{x,y} |\mathbb{E}_{\rho_j} \phi(G_x)\phi(G_y) - Q_j(x,y)| \le \Delta, \qquad \sum_j D_j + r \le A + \tau,$$

provided

$$N \ge \frac{C}{\tau}\left[Hk \log\left(1 + \frac{C(A/\tau+1)Hk}{\Delta}\right) + m\log(C(A/\tau+1)) + 1\right]. \tag{37}$$

Here z is the realization's actual output and $r = z^{\mathsf{T}} Q_H^{\dagger} z/2$. No symmetry of the event is needed.

Proof. For a reference feature law μ on $[-1,1]^m$, let $\Phi(h)$ list its k upper-triangular products and define

$$I_B(q) = \max_{\lambda \in [-B,B]^k} [\lambda \cdot q - \log \mathbb{E}_\mu e^{\lambda \cdot \Phi}].$$

A maximizer exists. Its exponential tilt ν has moment vector a; coordinatewise optimality on the box gives $\lambda_i(q_i - a_i) = B|q_i - a_i|$. Hence

$$I_B(q) = \mathrm{KL}(\nu\|\mu) + B\,\|q - a\|_1 \ge 0. \tag{38}$$

The same tilt of Gaussian preactivations has identical entropy and moments. The objective is $2k$-Lipschitz in the sup norm of λ. A mesh-η grid has at most $(1 + CB/\eta)^k$ points, and each fixed exponential likelihood ratio for N iid rows has mean one. Thus for empirical moments $\widehat{q}$,

$$\mathbb{E} e^{N I_B(\widehat{q})} \le e^{2Nk\eta}(1 + CB/\eta)^k.$$

This is uniform in the conditional reference. Iterate through layers. Given the hidden layers, the actual readout covariance is Q_H/N, so $\mathbb{E}[e^{(1-s)Nr} \mid \text{hidden}] \le s^{-m/2}$ for $0 < s < 1$. Writing I_j for the realized dual costs gives

$$\mathbb{E} e^{N[\sum I_j + (1-s)r]} \le e^{2NHk\eta}(1 + CB/\eta)^{Hk} s^{-m/2}.$$

Conditioning on E has relative entropy at most NA. Entropy duality bounds the conditional mean of $\sum I_j + (1-s)r$ by

$$A + 2Hk\eta + N^{-1}[Hk\log(1 + CB/\eta) + (m/2)\log(1/s)].$$

Take $s = \tau/[16(A+\tau)]$, $\eta = \tau/(64Hk)$, and $B = (A+\tau)/\Delta$. Under (37), some realization in E has weighted cost at most $A + \tau/8$, hence unweighted cost at most $A + \tau$, since all $I_j \ge 0$ and $(A+\tau)(1-s) = A + 15\tau/16$. Apply (38) at each layer; its moment error is at most $I_j/B \le \Delta$. □

Lemma B.9 (Consistent-path repair)**.** *A certificate of total cost at most B in Lemma B.8 has a genuine controlled path of cost at most B and output $\widehat{z}$ satisfying*

$$\max_x |\widehat{z}(x) - z(x)| \le \sqrt{2B}\, H \sqrt{m\Delta}\, e^{\sqrt{HB}}.$$

Proof. Two positive semidefinite m-point kernels differing entrywise by at most Δ have square-root feature embeddings with row distances at most $\sqrt{m\Delta}$. Their operator difference is at most $m\Delta$, and $\left\|K^{1/2} - K'^{1/2}\right\|_{\mathrm{op}} \le \sqrt{\|K - K'\|_{\mathrm{op}}}$. For the last inequality use $K \le K' + aI$, monotonicity of square root, and $\sqrt{K' + aI} \le \sqrt{K'} + \sqrt{a}I$, then reverse K, K'. Existing embeddings can be identified isometrically with these square-root embeddings, adjoining orthogonal complements.

At each step couple Gaussian rows for the genuine preceding kernel and the empirical reference in such a common feature space. Tilt the joint Gaussian by the certificate density on its original coordinate. The new marginal has no larger entropy. Equation (32) and the Lipschitz property multiply each row RMS error by at most $e^{\sqrt{D_j}}$. Identify the controlled feature moments with the next empirical kernel using the certificate error, at additional row distance $\sqrt{m\Delta}$. Gluing the shared feature spans gives $e_j \le e^{\sqrt{D_j}} e_{j-1} + \sqrt{m\Delta}$, $e_0 = 0$. Consequently $e_H \le H\sqrt{m\Delta} e^{\sqrt{HB}}$. Transfer the empirical readout vector of norm $\sqrt{2r}$ in the common space and project it onto the genuine final feature span. Its norm does not increase, and its output changes by at most $\sqrt{2r} e_H \le \sqrt{2B} e_H$. □

Theorem B.10 (Actual critical posterior circuit ceiling)**.** *Put $R_0 = d + n + L + 2$, and suppose $L \ge C \log R_0$. At the actual critical scale $\sigma^2 = 1/N$, for every integer $N \ge 2^{Cd} R_0^C$ and every consistently labeled sample of at most n inputs, the original posterior gives probability at least $31/32$ to a fixed full-cube class of functions having Boolean circuits of size $\mathrm{poly}(R_0)$. More precisely this class can be taken as $\{g : J_L(g) \le V_* + C \log^2 R_0/L\}$; its threshold depth is $O(\log R_0)$ and its Boolean depth is $O(\log^2 R_0)$.*

Proof. The memorizer $g_S = 1 - 2\sum_{p \text{ distinct negative sample input}} \mathbf{1}_{\{p\}}$ is Boolean, has the sample labels, and has Fourier variation at most $1 + 2n$, since each point indicator has Fourier variation one. Theorem B.5 gives a raw conditioning denominator at least e^{-Na_0}, where $a_0 = V_* + C_0 \log^2 R_0/L$, at all $N \ge R_0^{C_0}$ after enlarging constants.

Let E be the raw event of globally margined draws whose Boolean function satisfies $J_L(g) > a_0 + K_0/L$. This is measurable because there are finitely many Boolean functions on the cube. If $\mu_N(E) \ge e^{-N(a_0+1/L)}$, apply Lemma B.8 on the entire cube with $m = 2^d$, action parameter $a_0 + 1/L$, and $\tau = 1/L$. Choose Δ so Lemma B.9 has output error at most $\rho = [CL(a_0+1)]^{-1}$. It suffices that

$$\log(1/\Delta) \le C[d + \log(L(a_0+2)) + \sqrt{L(a_0+2)}]$$

for an appropriate sufficiently small tolerance. The required width in (37) is therefore at most $2^{Cd} R_0^C$. The repaired output keeps the selected signs; multiplying its readout by $(1-2\rho)^{-1}$ restores margin $1/2$ and adds $O(1/L)$ action. This realizes a function in E at action at most $a_0 + K_1/L$, contradicting $K_0 > K_1$.

Hence $\mu_N(E) < e^{-N(a_0+1/L)}$. Division by the sample-consistent denominator bounds posterior bad mass by $e^{-N/L} \le 1/32$ under the stated width threshold. Absorb K_0/L into the logarithmic excess. Theorem B.7 gives the claimed circuit bounds. If the rate infimum is not attained, choose a path within $1/H$; this changes the exponent $\sqrt{H\epsilon}$ only by a constant. Every bound is simultaneous over all samples of the specified size, and the class itself is sample independent. □

C A variant with every derivative globally bounded

The activation in Theorem 1.1 has global bounds on any prescribed finite number of derivatives. Its sufficiently high derivatives need not be bounded. This feature is not necessary for an exponential dependence on teacher depth: a smaller encoding amplitude gives one activation whose derivatives of *every* order are globally bounded. Amplifier weights of order d preserve the linear teacher depth.

Theorem C.1. *Fix a finite integer $r \ge 1$ and $\eta > 0$. There is one computable activation ϕ with all the shape and C^r-closeness properties of Theorem 1.1, such that every derivative of ϕ is globally bounded. The same circuit-hard targets f_d have globally unit-margin teachers with*

$$L_d = d + 4, \qquad \textit{active width } O(\log(d+2)), \qquad \textit{coefficient magnitude } O(d+1),$$

and polynomial coefficient bit length. For some $c > 0$, every sample of at most $\exp(cd)$ observations has posterior confidence-failure rate at least $139/1024$, at every fixed Gaussian exponent $\gamma \geq 1$ and every sufficiently large width $N \geq \exp(C_\gamma d)$. The fixed-width cold-limit conclusion holds at every width above the memorizer threshold (11).

Thus this variant retains the sample lower bound $\exp(\Omega(L_d))$. It does not assert a uniform analytic-strip bound, a finite-order Gevrey bound, or efficient activation evaluation.

C.1 The modified bands and derivative estimates

Fix a sufficiently large integer $M_0 \geq 24$, and set

$$m_d = M_0 + \lceil \log_2(d+2) \rceil, \qquad J_d = 2^{m_d+10}, \qquad \Delta_d = 2^{-d-4},$$
$$r_d = \Delta_d / J_d, \qquad c_{d,k} = -4d - \tfrac{1}{4} - \Delta_d k, \qquad \delta_d = \tau r_d 2^{-(m_d-3)d}.$$

In particular,

$$\delta_d = \tau 2^{-(m_d-2)d - m_d - 14}. \tag{39}$$

Use these parameters in (3)–(4), with a fixed sufficiently small positive rational τ. The truth tables and centers are otherwise unchanged.

For each fixed derivative order j, the Gaussian lattice estimate in Lemma 3.1 gives

$$\sup_{\mathbb{R}} |S_d^{(j)}| \leq C_j \tau (16 J_d)^{j-1} 2^{-(m_d-j-2)d}. \tag{40}$$

Here S_d denotes the whole dth band. This is summable in d for every fixed j, because $m_d = \log_2 d + O(1)$ and $J_d = O(d)$. Consequently every derivative is globally bounded. On each complex compact, the moving band centers and Gaussian suppression still give locally uniform convergence; hence S is entire. The explicit tails and the computable finite truth tables give computability.

The weighted first- and second-derivative estimates have band majorants

$$C\tau e^{8d} 2^{-(m_d-3)d}, \qquad C\tau J_d e^{8d} 2^{-(m_d-4)d},$$

respectively. Both are summable. The cubic factor at zero supplies the same extra powers of u as in Lemma 3.1. Therefore (6)–(7) hold, with fixed constants for this modified activation. A sufficiently small τ preserves oddness, slope one at zero, strict increase, global Lipschitz constant one, positive-half-line strict concavity, and saturation to ± 1. Equation (40) makes the prescribed C^r distance smaller than η after a further fixed reduction of τ. The constants bounding derivative j can grow rapidly with j; no uniform analytic-type estimate is used.

C.2 A growing finite-difference decoder

Use $h_d = \Delta_d/(2m_d)$ and the m_dth finite difference in (10). At $c_{d,k}$ the central bump contributes $\delta_d f_d(k)$. Every other query is at least h_d from every center in this band. Since $J_d = 2^{m_d+10}$, its leakage, multiplied by the coefficient sum $2^{m_d} = O(d)$, is $o(\delta_d)$.

For another band $e \neq d$, the logarithm of its amplitude ratio is $O(d \log d + e \log e)$, whereas Gaussian suppression has exponent at most $-c(d-e)^2 4^e J_e^2$. Split the sum into $e \leq d/2$, $d/2 < e < d$, and $e > d$. In the first range even the smallest fixed e gives a negative quadratic in d, dominating $d \log d$. The second has suppression at least $\exp(-c4^{d/2})$; in the third the amplitudes decrease and the Gaussian suppression is stronger still. Polynomial factors and the coefficient sum do not change these conclusions. Reflected bands are farther away.

For $m = m_d = O(\log d)$, the negative-half-line exponential series of tanh gives

$$\sup_{u \le -4d+1} |\tanh^{(m)} u| \le C 2^m e^{-8d+2}.$$

Indeed the differentiated series is bounded by $2^{m+1} \sum_{n \ge 1} n^m e^{-(8d-2)n}$, whose first term dominates uniformly for large d since $m = o(d)$. The tanh contribution to the finite difference is at most $C\Delta_d^m m^{-m} e^{-8d}$. Dividing by (39) bounds its relative size by

$$C\tau^{-1} 2^{-2d-3m+14} m^{-m} e^{-8d} = o(1).$$

We therefore retain the uniform decoding estimate $F_d(c_{d,k}) = \delta_d(f_d(k) + o(1))$.

The first layer has $m_d + 1$ neurons and evaluates these query points. The next layer uses binomial coefficients of magnitude at most $2^{m_d} = O(d)$. Set the amplifier gain

$$a_d = 2^{m_d - 2} = \Theta(d).$$

Exactly $d + 1$ scalar amplification layers give ideal final amplitude

$$\delta_d a_d^{d+1} = \tau 2^{-16} =: \rho.$$

Their ideal preactivation amplitudes form a geometric sequence with sum at most 2ρ, since $a_d \ge 2$. The estimate (8) therefore gives a uniform $O(\rho)$ relative-error bootstrap, independent of the growing gain. Choosing τ small keeps every actual preactivation below 2ρ; a fixed readout $2/\rho$ gives margin greater than one. Counting the query layer, combination layer, $d + 1$ amplifiers, and readout gives $L_d = d + 4$. The amplifier weights are still $O(d)$. This proves the teacher resource bounds. Rounding its coefficients to $O(d \log d)$ bits preserves the margin: the whole propagation gain is at most $\exp(O(d \log d))$.

C.3 Cold and fixed-gap posterior bounds

Let $a > 0$ be the hard-product exponent in the proof of Lemma 2.2. Since $L_d = d + 4$, the original fixed-gain memorizer applies. Set

$$q = \exp(a/100), \qquad \alpha = c_0(q - 1),$$

with the same sufficiently small constant c_0 as in Lemma 5.1. Both are fixed, and $q^{L_d} = \exp(ad/100 + O(1))$. The relative-error sum is still $O(\alpha/(q-1))$, hence uniformly small. The first scalar energy is at most one: its decaying factor is

$$q^{-2(L_d - 2)} = \exp(-ad/50 + O(1)),$$

whereas (12) gives $\log B_0 \le 3cd + O(\log d)$. Choose $c < a/1000$. The witness energy satisfies

$$K_d \le q^2 L_d + O(1). \tag{41}$$

The cold posterior argument remains valid at each fixed architecture. Arithmetic–geometric mean bounds its unit-margin representative product by

$$\exp\{ad/100 + O(1)\} \le e^{ad}$$

for all large d. This is the same hard-table threshold as before.

For the actual fixed-gap scales, Lemma 7.1 and Proposition 7.2 apply with the energy in (41). Since $L_d = d+4$ and $q^{L_d} = \exp(ad/100+O(1))$, the inverse perturbation tolerance remains $\exp(O(d))$ without further changes. The Gaussian bulk constant is polynomial in d, L_d, and the memorizer width in (11) is $\exp(O(d))$ uniformly over $n \le e^{cd}$. Thus every $N \ge \exp(C_\zeta d)$ meets the finite-scale condition (19), for each fixed gap $\zeta = \gamma - 1 > 0$. The posterior product bound remains below e^{ad}; the hard-table and Gibbs-to-confidence calculation gives the same failure rate 139/1024.

C.4 The critical endpoint

The modified activation satisfies the fixed-activation hypotheses of Appendix B: the same shape and weighted estimates, bounded derivatives through order five, and the strict scalar potential obtained by choosing τ sufficiently small. Therefore its finite-width target-mass and canonical-circuit results apply with constants depending only on this fixed activation.

For $L_d = d + 4$ and $n \le e^{cd}$, reducing c makes the depth condition $L_d \ge C \log(d + n + L_d)$ hold for all large d. A further reduction makes the posterior circuit ceiling $\mathrm{poly}(d + n + L_d)$ at most $2^{d/4}$. The width threshold $2^{Cd}(d + n + L_d)^C$ is still at most $\exp(C'd)$. The hard-table and confidence calculation is unchanged. This completes the proof of Theorem C.1.

C.5 Quantitative derivative bounds and numerical evaluation

The distinction between bounded derivatives of each order and a quantitative bound on their growth is important here.

Proposition C.2. *Suppose a real activation satisfies, for fixed constants $C, A \ge 1$ and finite $s \ge 1$,*

$$\sup_{u \in \mathbb{R}} |\phi^{(j)}(u)| \le C A^j (j!)^s \qquad (j \ge 0).$$

On dyadic inputs of bit length b in $[-R, R]$, it has nonuniform Boolean circuits approximating its value to error 2^{-p}, of size polynomial in $R + 1$, $p + 1$, and $b + 1$. The polynomial may depend on C, A, s. No efficient uniform procedure for computing the circuit's coefficients is assumed.

Proof. Replace s by the integer $s_0 = \lceil s \rceil$, and choose an integer upper bound for A. For an integer n, partition the interval into pieces of radius at most $h = (2An^{s_0-1})^{-1}$. Taylor expansion of degree $n - 1$ at the center of a piece has remainder at most

$$\frac{CA^n (n!)^{s_0}}{n!} h^n \le C(Ahn^{s_0-1})^n = C2^{-n}.$$

Take $n = O(p + \log C + 1)$, with enough fixed slack. There are $O((R + 1)n^{s_0-1})$ pieces. Store their centers and Taylor coefficients as nonuniform circuit constants. The coefficients have logarithmic magnitude $O(n \log(n + 1))$, and the local argument has magnitude at most one. Rounding each coefficient to absolute error $2^{-p}/(8n)$ controls the total coefficient error. Polynomially many additional bits suffice for the dyadic arithmetic, interval selection, and evaluation of each polynomial. All sizes are polynomial in the displayed resources. □

This includes a global finite-order Gevrey bound. A bounded holomorphic extension to a fixed-width strip is another sufficient condition, by Cauchy estimates. Such conditions cannot hold for the constructed activation: simulating its polynomial-resource teacher at precision $O(d \log d)$ would give polynomial-size Boolean circuits for the circuit-hard f_d. Thus the new variant still contains computationally expensive information in its activation primitive. It is not a counterexample for a standard efficiently evaluable activation or for exact tanh.